Spectrometer Manager SpecMan

a powerful control system for EPR spectrometers

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In modern commercial EPR spectrometers, the controlling software is tightly integrated with the hardware. This integration has many advantages for the vendors. For example, it secures the vendors complete control over the product for which they have ultimate responsibility. However, a growing number of EPR laboratories, especially those that develop instrumentation, employ homebuilt equipment for which no commercial software exists. These research groups have developed various programs to control their instrumentation. The programs roughly fall into two groups. The first group consists of dedicated software with a fixed structure tailored to a particular configuration of instruments. They are typically written in languages like C++, Basic or Pascal. The second group of programs is written using rapid application development platforms, e.g., LabView. There exist also freeware development environments for EPR instrumentation (e.g., fsc2 [1]). The applications of the second group are typically more flexible and expandable. However, for both groups, substantial changes to the software are necessary when performing experiments with changed instrumental setup or when migrating to different instruments.

SpecMan combines the advantages of both groups and provides increased flexibility and portability compared to either of the above groups. It is a stand-alone Microsoft Windows™ application, written in C++. Nevertheless, the combination of easily editable instrumental configuration files and a built-in scripting language for pulse sequences makes it truly flexible. SpecMan is designed to perform continuous-wave and arbitrary pulse EPR experiments, with the emphasis on the latter. The detailed structure of the program is described elsewhere [2]. A simple but powerful Pulse Programming Language PPL (inspired by the language developed by Jaap Shane et al. [3]) allows for an intuitive representation of virtually any pulse sequence. The innovative Graphical User Interface dramatically simplifies the control of an experiment. A robust kernel executes experiments at the highest speed. A large device driver library is provided, containing drivers for all devices commonly employed in pulse EPR spectrometers.

SpecMan is tailored to the needs of pulse EPR and has the number of unique features. These include random acquisition of parameters (instead of left-to-right sweeps), multiple signal detection during the pulse sequence, execution of several pulse sequences in one run and a number of data protection mechanisms. These features provide a powerful environment for a wide variety of the pulse EPR experiments.

Functionally, SpecMan has two components, a client where experimental parameters are set and data are displayed during acquisition, and a server that controls the hardware and collects data via the hardware interface (GPIB bus cards, pulse generators, A/D converters, etc.). These two components of SpecMan communicate via an internal messaging protocol based in TCP/IP. This way, the client and the server can interact in a seamless fashion even when running on separate machines. The development of SpecMan network capabilities is currently in progress.

One of the problems that have to be solved is the limited network channel bandwidth. In a typical pulsed EPR experiment, a huge amount of data is generated rapidly. When SpecMan operates on a single computer, the data can be directly written into the memory by the server and displayed simultaneously by the client. When SpecMan operates on two different computers, passing the entire data across a TCP/IP channel in real time is not possible. For example, an experiment with a repetition time of 1 ms, a time trace of 5000 points and two-channel acquisition with 8-bit resolution will generate 10 MB of data per second. This dataflow can block even the fastest network. Hence, during the experiment, the server sends only a subset of the data asynchronously to the client for display purpose. Assuming that the computer screen is roughly 1000 pixels wide, no more than 1000 data points have to be passed over the network to the client computer during the experiment. Once the experiment is completed, the full data set is sent to the client. In this way, SpecMan can both collect the data and inform the user of the progress of the experiment at the highest possible speed.

A number of research groups around the world already take advantage of using SpecMan as spectrometer control software. The new network capability will extend the applicability of the program e.g. to the case of performing experiments on dangerous samples or noisy environments.

At the beginning of this year we lost Prof. Arthur Schweiger, who played a key role in the development of SpecMan. SpecMan is distributed through Scientific Software Services. Questions should be directed to Reef Morse at reef@scientific-software.com.

† Passed away on 4th January 2006

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